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ABSTRACT

Doping, which has been extensively studied in the past century, is essential for tuning the electrical conductivity of semiconductors. However, the effect of doping on the thermal conductivity of semiconductors has been much less studied despite the growing importance of electronics cooling. A clear understanding of defect–phonon scatterings remains elusive due to difficulties in well-controlled growth of crystals and advanced thermal and structural characterizations. Herein, by combining advanced characterizations and high-quality growth of cubic silicon carbide (3C-SiC) single crystals with well-controlled boron doping, we experimentally observe an extremely strong defect–phonon scattering intensity. This effect exceeds the prediction of the classic mass-difference model by tens of times in magnitude. At the doping level of 10^{19} cm^{-3}, a 50% reduction of thermal conductivity was achieved—the highest reductions reported for common semiconductors. The measured thermal conductivity of B-doped 3C-SiC matches first-principles predictions with excellent agreement, demonstrating the critical role of low-frequency phonon resonant scattering. The influence of strain gradients on thermal transport was excluded by local vibrational spectra scanning through atomic-resolution phonon-level electron energy loss spectroscopy. Our findings not only shed light on the fundamental understanding of defect–phonon interactions but will also impact applications such as thermal management of electronics.

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I. INTRODUCTION

Doping, which intentionally introduces specific impurity atoms to supply additional electrons or holes, has been widely studied and extensively used to modulate carrier concentrations in semiconductors, facilitating the desired electrical properties for logic, memory, power, and optoelectronic devices.[1,2] Through precise control of carrier concentrations, doping has become a fundamental process in modern microelectronics. However, as electronic devices continue to shrink in size and increase in power density, thermal dissipation has emerged as a critical bottleneck, particularly for three-dimensional integrated circuits and high-power/high-frequency wide-bandgap semiconductor devices.[3,4] The thermal conductivity of semiconductors is fundamentally critical for the thermal management of micro- and nanostructured electronic devices, directly impacting their stability, reliability, and lifespan. While the effects of doping on electrical properties have been extensively studied, a clear understanding of doping-induced point defects and the reduction in thermal conductivity in semiconductors remains elusive due to the difficulties in sample growth and advanced thermal and structural characterizations.

According to classic kinetic theory, the thermal conductivity k of a semiconductor is given by $k = \sum_i \frac{1}{3} C_i v_i^2 \tau_i$, where C_i, v_i, τ_i represent the heat capacity, group velocity, and relaxation time between two phonon scattering events for a specific polarization mode i.[5] The total relaxation time τ_i is determined by contributions from different scattering mechanisms—such as phonon–phonon, phonon–defect, and phonon–electron interactions—following Matthiessen's rule: $\tau^{-1} = \sum_n \tau_n^{-1}$.[6] In conventional doped semiconductors, substitutional defects are generally considered to induce Rayleigh scattering. Within the mass perturbation theory, the phonon–defect scattering rate is proportional to the square of the mass difference between the host atom

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and the substituted atom$(\Delta M/M)^2$.[7] This scattering model has been successfully applied to predict the thermal conductivity of semiconductors such as 4H-SiC, 6H-SiC, and GaN.[8,9] However, Rayleigh scattering under the mass perturbation assumption may not fully account for all phonon–defect interactions. A thorough investigation of these effects is essential for a comprehensive understanding of thermal conduction in doped semiconductors.

In the 1960s, Rosenberg and Sujak observed an anomalously large reduction in thermal conductivity at low temperatures in hydrated zinc sulfate crystals doped with Fe^{2+} ions compared to those doped with Mg^{2+} ions, despite Mg exhibiting a much larger mass difference from the host Zn atom.[10] They attributed it to phonon–spin scattering near a specific frequency, where the energy of a phonon induces the transition between different spin energy levels. Similarly, Pohl observed the interaction between phonons and NO_2^- in a nonparamagnetic KCl crystal that led to a "dip" in thermal conductivity at low temperature, which cannot be explained by conventional Rayleigh scattering theory.[11] They refer to this type of non-Rayleigh phonon scattering, involving interaction between spin and phonons near a certain frequency, as resonant phonon scattering, which significantly decreases the thermal conductivity of the host material. A similar scattering phenomenon has also been observed in thermoelectric materials, where weakly bound filler atoms inside cage-like structures induce resonant scattering.[12]

While early studies by Pohl, Rosenberg, and others primarily focused on dielectric materials, the effect of phonon resonant scattering in doped semiconductors is of great importance. Several theoretical and experimental investigations,[13–18] particularly on cubic silicon carbide (3C-SiC)—a key wide-bandgap semiconductor which possesses the second-highest thermal conductivity among large crystals (only surpassed by diamond)—have been reported.[14,15,17] A 3C-SiC thin film has been shown to have significantly reduced thermal conductivity compared to high-purity 3C-SiC bulk crystal. However, the effects of impurities and boundaries are not separated, making it difficult to understand the impact of defect–phonon scatterings alone.[17] The thermal conductivity of polycrystalline 3C-SiC ceramic materials, which are doped with nitrogen or co-doped with nitrogen and boron, has also been studied.[19] Nevertheless, previous theoretical and experimental results are controversial. For instance, in the theoretical study,[15] the grain size used for fitting the theoretically calculated thermal conductivity ($\sim$0.8 μm) to the measured data of the 3C-SiC ceramic was nearly an order of magnitude smaller than the measured actual grain size (1–10 μm).[19] The strong grain boundary–phonon scatterings may compensate or couple with the defect–phonon scatterings. Additionally, Ivanova *et al.*'s report on nitrogen-boron co-doped 3C-SiC ceramics[19] is limited for defect analysis since it only provides carrier concentrations—a parameter that generally differs from defect concentrations, even in single-element-doped semiconductors. In the specific case of nitrogen-boron co-doped 3C-SiC, the carrier concentration would only represent the difference between activated nitrogen and boron dopants. However, as only a small fraction of boron dopants become activated at room temperature, substantial deviations in dopant concentrations are caused by relying solely on carrier concentration.[20] Thus, separate quantification of nitrogen and boron defect concentrations requires additional advanced dopant characterization. Moreover, other structural imperfections may also scatter phonons due to the low quality of the 3C-SiC ceramic samples.

The limited availability of high-quality 3C-SiC single crystals with well-controlled doping levels, along with insufficient advanced structural and thermal characterizations, hinders the fundamental understanding of phonon scattering mechanisms—particularly the theory of resonant phonon scattering in 3C-SiC. In addition to well-designed sample growth, a detailed characterization of dopant distribution and its effect on phonon scattering is also crucial for a clear understanding of thermal conduction in doped semiconductors,[21] as dopant clusters and non-uniform distribution may show up.

Here, we present advanced thermal and structural characterizations on high-quality boron-doped 3C-SiC single crystals to investigate defect–phonon scattering effects on thermal conductivity. By utilizing low-temperature chemical vapor deposition (LT-CVD) and time-domain thermoreflectance (TDTR), we synthesize high-quality 3C-SiC single-crystal samples and measure their thermal conductivity as a function of temperature and doping concentration. The crystal quality was comprehensively characterized by scanning transmission electron microscopy (STEM), Raman spectroscopy, and electron backscatter diffraction (EBSD). Secondary ion mass spectrometry (SIMS) and atom probe tomography (APT) are employed to quantify the concentrations and elemental distribution of dopants. Moreover, the significant lattice mismatch between Si and 3C-SiC can generate considerable tensile stress gradients,[22,23] which may enhance phonon scattering by broadening the phonon dispersion spectrum.[24] To investigate this phenomenon, atomic-resolution STEM-EELS[25–28] was employed to measure the energy shift of phonon modes across the film thickness to reveal nonhomogeneous stress.[24] This comprehensive approach enables direct comparison between experimental results and theoretical models, thereby facilitating in-depth analysis of the resonant phonon scattering mechanism and providing guidance for thermal design in relevant electronic devices.

II. RESULTS AND DISCUSSION

A. Structural characterizations

Four 3C-SiC single crystals with intentionally controlled boron doping are grown by LT-CVD. More details about the growth process can be found in Sec. IV and Ref. 17. The boron concentrations within the four crystals are determined by SIMS: $\sim 1 \times 10^{16}\,cm^{-3}$ (non-doped, the detect limit of the SIMS technique), 6×10^{18}, 1.5×10^{19}, and $4 \times 10^{19}\,cm^{-3}$. Other unintentional dopants, such as nitrogen and oxygen, are also measured and found to have low concentrations, which are not expected to significantly affect the thermal conductivity. The depth profiles of the respective dopants for all four sample sets are presented in Fig. S1 of the supplementary material. The sample thicknesses are measured using the picosecond acoustic technique in the TDTR system, and the result is shown in Fig. 1(a), with the inset illustrating the principle. The thickness measured by the picosecond acoustic technique ($\sim$3.5 μm) is much larger than the thermal penetration depth in the TDTR measurements (further explanation can be found in the supplementary material), which supports that the samples are thermally thick and can be treated as bulk crystals. The large-area EBSD maps presented in Figs. S2 and S3 demonstrate the single (111) orientation of both the unintentionally doped and boron-doped ($6 \times 10^{18}\,cm^{-3}$) 3C-SiC samples. The absence of grain boundaries in these maps further confirms the high crystalline quality and monocrystalline nature of the samples.

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Figure 1(b) shows the atomic-scale high-angle annular dark field (HAADF)-STEM image of the high-quality 3C-SiC crystal, with the interplanar spacings of the lattice planes agreeing well with Refs. 29 and 30. The inset image shows the fast Fourier transform (FFT) of the HAADF-STEM image, indicating the single-crystal nature of the selected area. Large-area TEM and STEM images of the unintentionally and heavily boron-doped (1.5×10^{19} cm^{-3}) samples are provided in Fig. S4 for more comprehensive structural analysis. In Fig. S4(a), the folds observed in both the 3C-SiC layer and the Si substrate are artifacts caused by the thin sample, rather than intrinsic material defects. Figure S4(b) clearly reveals the 3C-SiC/Si heterointerface, exhibiting a high density of threading dislocations and structural defects resulting from the substantial lattice mismatch between 3C-SiC and Si.[23] However, due to the limited thermal penetration depth, our thermal measurements are not sensitive to the low-quality interfacial regions of the sample. Figures S4(c) and S4(d) show the HAADF-STEM images of the atomic crystal structure of the two samples in a larger area and their corresponding FFT images.

To further verify the sample's crystalline quality, we performed Raman spectroscopy analysis, which provides sensitive detection of structural defects and strain in the material. The Raman spectra of both the 3C-SiC samples in this work (upper curve) and a free-standing 3C-SiC bulk sample (lower curve)[17] are shown in Fig. 1(c). The shifts and full width at half maximum (FWHM) values of the Raman peaks of both samples are consistent, indicating the high quality of the films. The broadening at the base of the second Raman peak is attributed to the Si substrate.[31] The longitudinal optical (LO) and transverse optical (TO) phonon frequencies at Γ point which correspond to the Raman peak positions, are illustrated in Fig. 1(d). Good agreement is observed with the calculated 3C-SiC phonon dispersion curve.[32]

Although Raman spectroscopy indicates that the total stress in the overall sample is negligible,[33,34] the substantial lattice mismatch

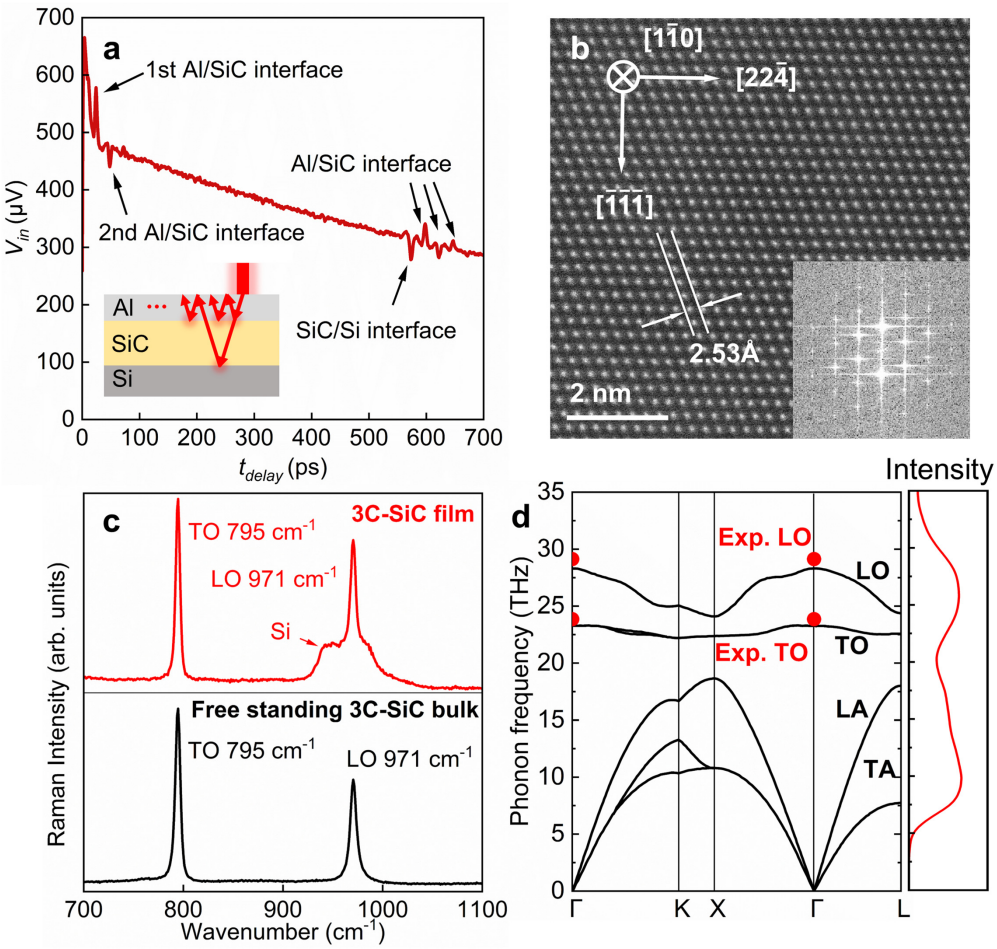


FIG. 1. Structural characterizations of the 3C-SiC sample. (a) Picosecond acoustic signal of the sample, with the inset illustrating the measurement principle. (b) HAADF-STEM image of the 3C-SiC crystal layer along [1$\bar{1}$0] axis, with the inset showing the FFT of the STEM image. (c) Raman spectra of the 3C-SiC film in this study (upper curve) and a free-standing 3C-SiC bulk (lower curve). (d) The TO and LO phonon frequencies at Γ point correspond to the Raman results, compared with the calculated phonon dispersion curve.[32] Furthermore, the phonon energy spectra obtained by STEM-EELS (right column) demonstrate strong consistency with both the Raman experimental data and theoretical predictions. Reproduced with permission from Wang *et al.*, Phys. Rev. Mater. **1**, 034601 (2017). Copyright 2017 American Physical Society.[32]

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between Si and 3C-SiC is expected to generate significant tensile stress near the 3C-SiC/Si interface that gradually relaxes toward the surface. This inhomogeneous stress distribution has been demonstrated to enhance phonon–phonon scattering through phonon dispersion spectrum broadening, resulting in a more pronounced thermal conductivity reduction compared to uniformly distributed stress of equivalent magnitude,[24] despite being below Raman's detection threshold.

To characterize this stress gradient, we employed atomic-resolution STEM-EELS with millielectron-volt (meV) energy resolution to probe local lattice variations at different depths, with the experimental schematic shown in Fig. S5(a). Under vacuum conditions, sub-10 meV energy resolution [Figs. S5(b) and S5(c)] can be achieved, ensuring high measurement accuracy. The measurement results are shown in Figs. 1(d) and 2. Figure 2(a) shows the electron energy loss spectroscopy (EELS) scanning configuration, with a 3600 nm total scan length using 200 nm steps, intentionally offset from the surface to avoid surface phonon polariton (SPhP) effects.[35] The right column of Fig. 1(d) displays representative spectra where the acoustic (30–60 meV) and optical (100–120 meV) phonon modes show excellent agreement with both Raman data and theoretical predictions, confirming the measurement's reliability. Further validation comes from Figs. 3(b) and 3(c), which compare the scanning results with the DFT-calculated phonon density of states (PDOS) of 3C-SiC.[32] The strong correlation in phonon mode positions not only verifies the measurement accuracy but also confirms the high quality of the 3C-SiC sample. Analysis of Fig. 2(c) reveals stable phonon energies at depths of 3000–3200 nm, indicating stress-free conditions in this region. This conclusion is supported by the absence of significant Raman peak shifts in Fig. 1(d). The observed stability suggests the sample remains stress-free beyond approximately 3000 nm from the surface.

At shallower depths, the transverse acoustic (TA) mode exhibits a small blue shift starting from 3000 nm, while the TO mode shows a corresponding red shift beginning at 3200 nm. These energy shifts follow the same trend reported in previous studies of Si-nanowires under tensile stress,[24] demonstrating the existence of a weak local tensile stress gradient originating at the interface and extending about 600 nm into the film. Such a phenomenon can be more clearly observed in Fig. S6. This gradient results from lattice mismatch-induced expansion of atomic spacing near the interface that gradually recovers to normal structural parameters, creating an inhomogeneous stress region, as highlighted by the blue and white arrows in Figs. 2(c) and 2(d). Importantly, this stress gradient does not affect the thermal measurement results because the thermal penetration depth in this study (less than 2700 nm) lies entirely within the stress-free region. Consequently, the near-interface inhomogeneously stressed region remains undetected in thermal characterization.

The spatial distribution of boron dopants was analyzed using APT, which is capable of mapping the three-dimensional distribution of specific elements at atomic resolution.[36–39] The sample for APT measurements is fabricated into a needle-shaped tip using a focused ion beam (FIB) system to enhance the localized electric field near the tip, thereby facilitating the evaporation of surface atoms [Fig. 3(a)]. More details about APT measurements can be found in the materials and methods section. The atomic maps of silicon and carbon are shown in Figs. 2(c) and 3(b), respectively. These two elements, which form the host material SiC, are uniformly distributed. The atomic map of boron [Fig. 2(d)] reveals a wave-like distribution at the atomic scale, different from the uniform distributions of silicon and carbon. However, no B clustering is observed, and SIMS measurements show uniform distributions at a larger scale along the depth direction as

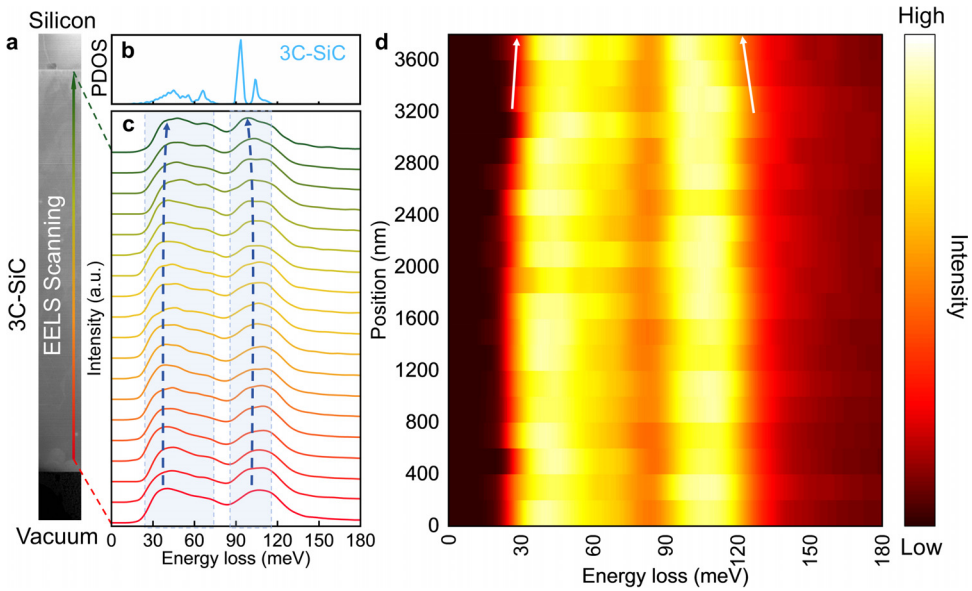


FIG. 2. Atomic-scale strain analysis using meV-resolution STEM-EELS. (a) HAADF image showing the sample cross section with superimposed line-scan trajectory. (b) Reference DFT-calculated phonon density of states (DOS) for 3C-SiC. (c) Depth-dependent phonon energy spectra acquired with a 200 nm step size, where blue-shaded regions highlight the characteristic acoustic and optical phonon windows of 3C-SiC. Dark blue arrows indicate observed phonon energy shifts. (d) Spatial mapping of acoustic and optical phonon mode energies along the scan direction, revealing stable phonon energies up to 3000 nm depth (white arrows mark energy shifts in the final 600 nm region).

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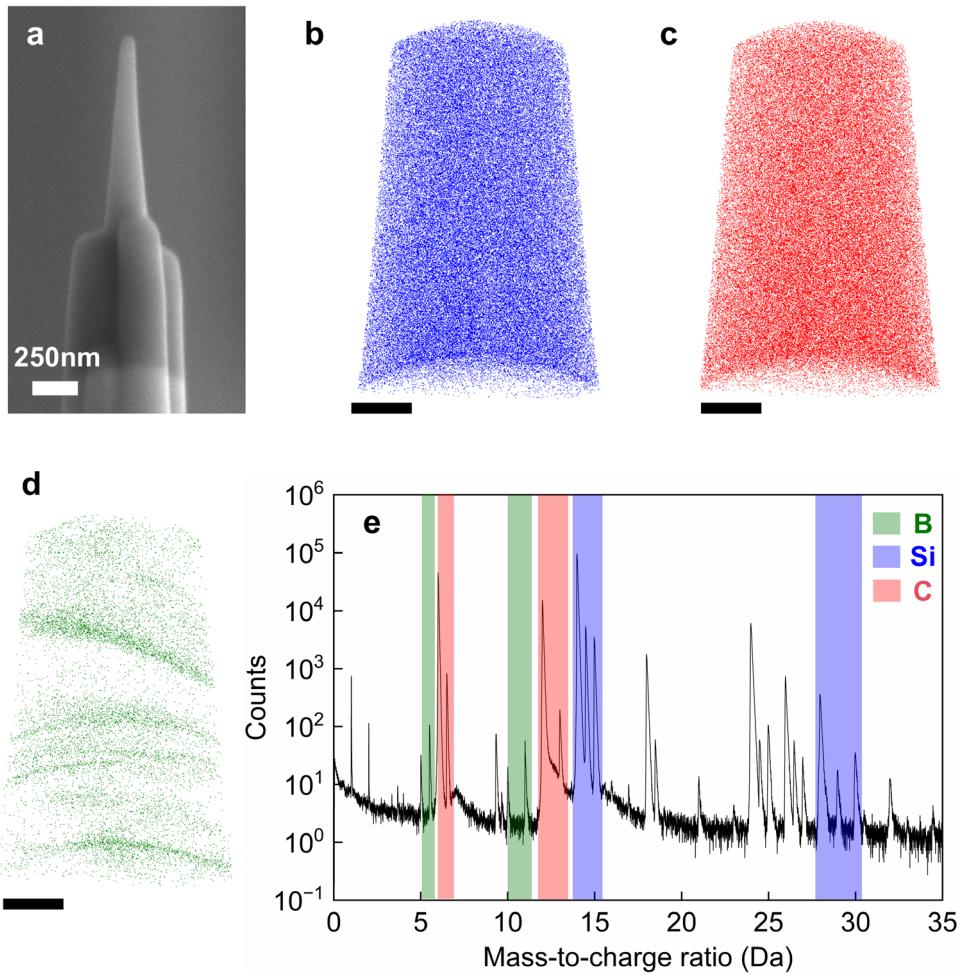


FIG. 3. Atomic-scale element distributions in 3C-SiC revealed by APT. (a) Needle-shaped sample prepared for APT characterization. (b)–(d) Atomic distribution maps of silicon (b), carbon (c), and boron (d), the scale bar is 20 nm. (e) Mass spectrum acquired via APT, with peak positions corresponding to boron, silicon, and carbon.

shown in Fig. S1. Therefore, we expect that the non-uniform B distribution has minimal impact on the thermal conductivity of 3C-SiC. The mass spectrum acquired after adjacent averaging smoothing via APT is shown in Fig. 2(e), with the three elements highlighted in different colors. Variations in the mass-to-charge peaks of the same element arise due to the presence of isotopes and doubly charged ions. The original data without smoothing are provided in Fig. S7. In Fig. S8, the comprehensive distribution of all three elements is shown.

B. Thermal characterizations

The thermal conductivity of the 3C-SiC single crystals was measured by the TDTR technique. More details about TDTR can be found in the experimental/methods section, Figs. S7–S8, and Ref. 40. Figure 4(a) presents the TDTR fitting data of the four samples with different doping levels. As the boron doping concentration increases, the TDTR ratio decreases, indicating a reduction in the thermal conductivity of 3C-SiC single crystals. The mechanism behind this strong reduction in thermal conductivity is illustrated in Fig. 4(b). When B atoms substitute for C atoms, the bond lengths between the defect and surrounding Si atoms undergo slight modifications, leading to a transition from tetrahedral to threefold symmetry.[14,15] This transition introduces a strong perturbation in the interatomic force constants (IFC), as exemplified by the one-dimensional potential energy curve shown in Fig. 4(b). The geometric distortion of the central atom causes the overlap of two potential minima, resulting in a significant change in the second derivative of the potential energy, which directly influences the IFC.[20] This strong perturbation induces an anomalous peak in the phonon–defect scattering rate at low frequency. Thus, the thermal conductivity was dramatically suppressed because low frequency acoustic phonons carry most of the heat. As a result, phonons near this resonant frequency experience significantly stronger scattering with B defects, leading to a much larger thermal conductivity reduction than that predicted by mass differences alone.

Figure 4(c) compares our experimental results with previous experimental and theoretical studies. The excellent agreement between the measured and calculated thermal conductivity of the un-doped single crystal shows that our samples have high-quality and are thermally

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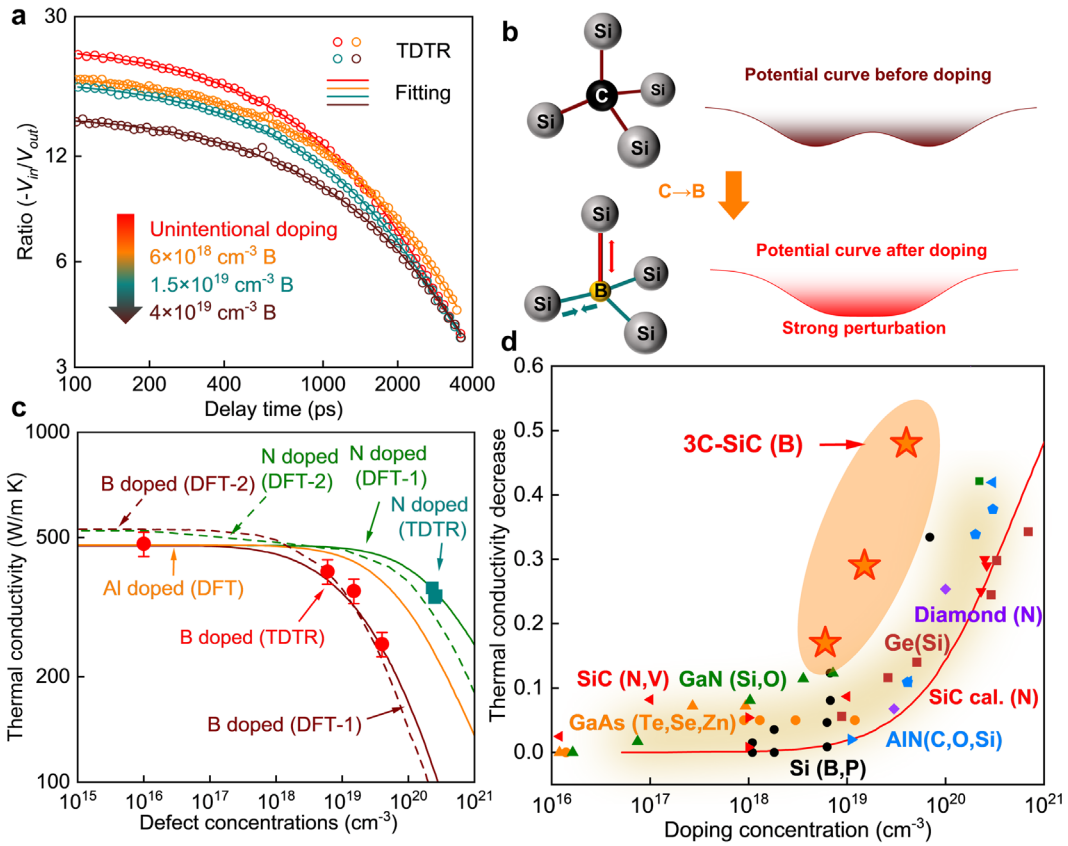


FIG. 4. Thermal transport in 3C-SiC single crystals at different boron doping concentrations. (a) TDTR experimental data and corresponding best-fit results for different doping concentrations. (b) Schematic illustration of resonant phonon scattering around a B defect in 3C-SiC: the substitution of a B atom for a C atom induces structural asymmetry, significantly perturbing the local potential field. (c) Comparison of experimental and theoretical studies on the relationship between thermal conductivity and doping concentrations in 3C-SiC. (d) Summary of normalized thermal conductivity reductions in different common semiconductors under various doping conditions.

thick in the TDTR measurements. Moreover, the measured thermal conductivity of the doped crystals also aligns well with theoretical predictions, providing strong evidence for resonant phonon scattering induced by intentional B doping. This scattering phenomenon significantly affects the thermal conductivity of 3C-SiC, leading to an approximately 50% reduction at a doping concentration of $\sim 4 \times 10^{19}\,\mathrm{cm}^{-3}$. Additionally, the theoretical thermal conductivities for varying N concentrations, as predicted by Katre *et al.*[15] and Pang *et al.*,[20] are also shown in Fig. 4(c). At a doping concentration of $1 \times 10^{19}\,\mathrm{cm}^{-3}$, N impurity has a negligible effect on thermal conductivity, whereas B doping results in a thermal conductivity reduction of approximately 25%. The experimentally measured thermal conductivity of nitrogen-doped 3C-SiC single crystals is also included in Fig. 4(c) (Ref. 41) for comparison, which agrees well with theoretical predictions. To achieve a similar reduction in thermal conductivity of 3C-SiC, the nitrogen doping concentrations need to be tens of times in magnitude higher than the boron doping. This occurs because the scattering mechanisms in the N-doped 3C-SiC are mass difference-induced phonon–defect scattering and phonon–electron scattering (in Pang *et al.*[20] and Wang *et al.*[32]) without the contribution of resonant phonon scattering. If the symmetry transition effect were ignored for B-doped 3C-SiC, the impact on thermal conductivity would be close to that of Al atoms substituting for Si atoms, corresponding to the effect of pure mass difference-induced scattering. The orange solid line in Fig. 4(c) represents this scenario, showing only a ~7% reduction in thermal conductivity at a doping concentration of $1 \times 10^{19}\,\mathrm{cm}^{-3}$, which is significantly weaker than the B-doped case.

This study reports the experimental observation of an exceptionally strong thermal conductivity reduction caused by boron doping in common semiconductor materials. Figure 4(d) shows the normalized thermal conductivity reduction in various semiconductors doped with different impurities, including commonly used semiconductors across a broad range of bandgaps, such as silicon,[42] germanium,[43] gallium arsenide,[44,45] silicon carbide,[41,46] gallium nitride,[47,48] aluminum nitride,[48–51] and diamond.[52] The doped elements for each material are also shown. The solid lines represent theoretical predictions for silicon carbide[20] doped with nitrogen, which is in good consistency with experiment.[41] It is clearly evident that B-doped 3C-SiC exhibits the largest percentage reduction in thermal conductivity across the entire studied concentration range.

We also notice that there are large variations and controversy for the experimental data in the literature. Most of the samples were not

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comprehensively characterized to obtain the detailed information, such as crystal quality, dopant distributions, stress distribution, and other structural imperfections. Potential contributions from grain boundaries, dislocations, and unintentional defects—along with their spatial distributions—could also contribute to the thermal conductivity reduction, but comprehensive studies characterizing and analyzing these factors are currently unavailable in the literature. Moreover, systematic and accurate thermal conductivity measurements under controlled conditions are still lacking. These make it difficult to fully capture the impact of impurity doping alone on thermal conductivity independently, underscoring the importance of a thorough examination of doping concentration and dopant atom distribution in this study.

Figure 5(a) compares the measured thermal conductivity of B-doped 3C-SiC from this study with literature data for N and B co-doped 3C-SiC reported by Ivanova *et al.*,[19] heavily N-doped 3C-SiC from Huang *et al.*,[41] and DFT calculations from Katre *et al.*[15] The unintentionally doped 3C-SiC single crystal has a thermal conductivity that excellently matches the DFT-calculated thermal conductivity value of a perfect 3C-SiC single crystal, highlighting again the high quality of the samples in this work. Also, it shows that the effects of other unintentionally doped impurities on the thermal conductivity of the 3C-SiC single crystals in this study are negligible. In comparison, the co-doped polycrystalline 3C-SiC ceramic samples from Ivanova *et al.*[19] exhibit much lower thermal conductivity, attributed to intense phonon scattering at grain boundaries and by other structural imperfections.

To systematically investigate thermal transport in B-doped 3C-SiC, we also measured the temperature-dependent thermal conductivity of the four samples from 300 to 700 K, as shown in Figs. 5(b) and 5(c). Figure 5(b) presents the temperature-dependent thermal conductivities for samples with the lowest and highest doping concentrations. The results of unintentionally doped 3C-SiC show excellent agreement with the DFT prediction of Katre *et al.*[15] over the entire temperature range. Figure 5(c) compares our experimental results of the other two doping levels with the theoretical predictions of Pang *et al.*[20] At similar doping concentrations (1.5×10^{19} and 0.6×10^{19} cm^{-3} vs 1.0×10^{19} cm^{-3}), our measurements align well with the theoretical

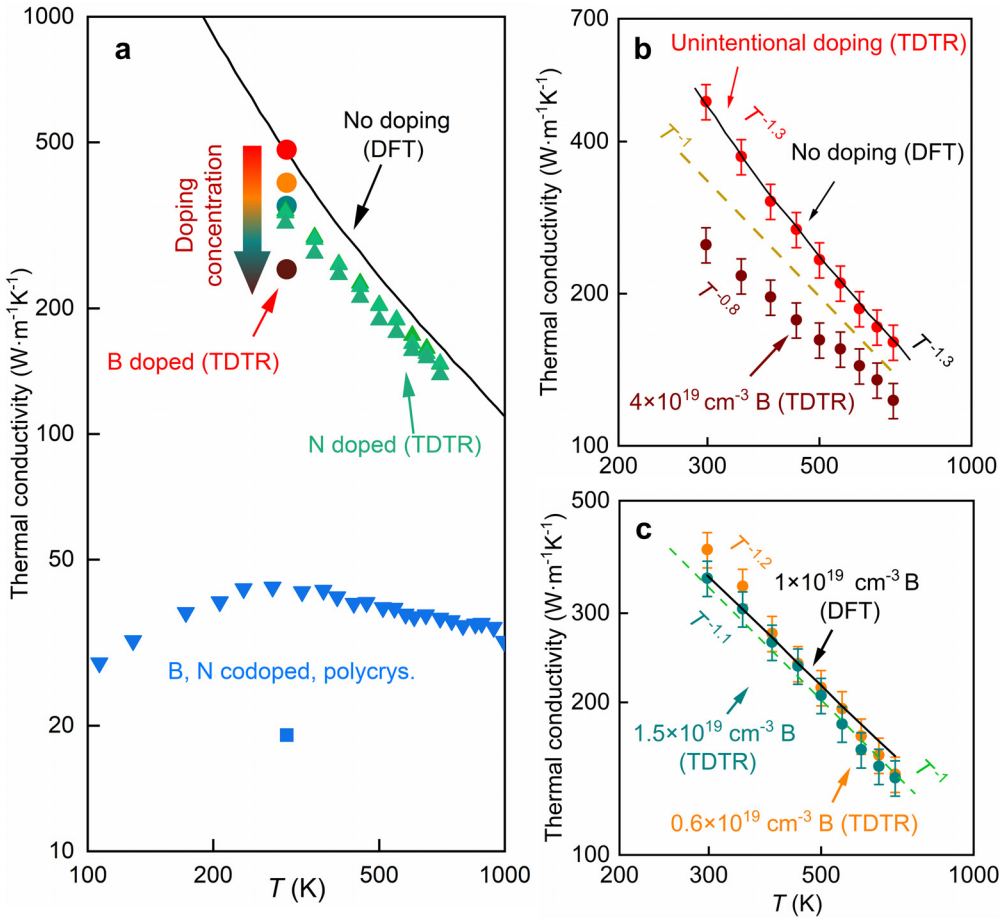


FIG. 5. Temperature dependence of thermal conduction in 3C-SiC single crystals. (a) Comparison of the measured thermal conductivity in this study with literature data from Ivanova *et al.*[19] and Huang *et al.*,[41] alongside DFT calculations from Katre *et al.*[15] (b) Temperature-dependent thermal conductivity of unintentionally doped and 4×10^{19} cm^{-3} B-doped 3C-SiC, compared with the calculation results from Katre *et al.*[15] (c) Temperature-dependent thermal conductivity of 3C-SiC doped with 6×10^{18} and 1.5×10^{19} cm^{-3} B, alongside the theoretical prediction of 1×10^{19} cm^{-3} B-doped from Pang *et al.*[20]

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predictions up to 500 K. All measured data follow logarithmic linearity, converging at elevated temperatures as Umklapp phonon–phonon scatterings dominate over phonon–defect scatterings.[53] Additionally, as the doping concentration increases, the temperature dependence of thermal conductivity weakens, as reflected in a reduced absolute slope in logarithmic scale, the T^{-1} line is also shown as a reference in Figs. 5(b) and 5(c) as a baseline. This trend indicates that phonon–defect scattering progressively dominates over phonon–phonon scattering at higher doping levels at low to medium temperatures. Consequently, as defect concentration continues to rise, the effect of temperature on thermal conductivity may become even weaker, resulting in flatter lines.

It is worth emphasizing that boron-doped 3C-SiC exhibits thermal transport characteristics that are qualitatively distinct from those of high-purity or conventionally doped 3C-SiC, as demonstrated by the present experimental and theoretical results. Based on these findings, several implications for thermal management in electronic devices incorporating boron-doped 3C-SiC can be drawn:

(i) When 3C-SiC is used as a device substrate, it often constitutes the primary pathway for heat dissipation; therefore, its chemical purity should be carefully controlled, particularly with respect to the minimization of boron impurities, since the impact of boron on thermal conductivity in 3C-SiC is significantly stronger than that typically observed for dopants in other material systems at comparable concentrations.

(ii) When boron doping is unavoidable in 3C-SiC, such as in the case of achieving *p*-type doping for the device's function, the dopant concentration should be carefully considered. According to the present results, when the boron concentration exceeds approximately 5×10^{19} cm^{-3}, the thermal conductivity of 3C-SiC can be reduced by more than 50%, which has a significant impact on the device's peak channel temperature and operational lifetime. This degradation becomes even more pronounced when B-doped 3C-SiC is located in regions of high local heat flux, as it strongly hinders the initial lateral spreading of heat.

(iii) Although boron-doped 3C-SiC exhibits a reduced thermal conductivity, its temperature dependence is noticeably weaker than that of high-purity 3C-SiC. This behavior should be taken into account in device-level thermal simulations, where temperature-dependent material properties are required to accurately predict the device temperature distribution.

III. CONCLUSION

In summary, this study presents the experimental observation of extremely strong defect-phonon scatterings in boron-doped single-crystal 3C-SiC, showcasing the strongest thermal conductivity reduction—nearly 50% at a doping concentration of 4×10^{19} cm^{-3}—observed among common semiconductors. Using the advanced TDTR method, we systematically measured the thermal conductivities across different doping levels, with results in excellent agreement with resonant phonon scattering theory. Moreover, temperature-dependent measurements further confirm room temperature observations. Advanced structural characterizations, including HAADF-STEM, Raman spectroscopy, and EBSD, confirmed the high quality of the 3C-SiC single crystal and enabled the exclusion of other structural imperfections from boron doping. SIMS and APT provided detailed analyses of the nanoscale dopant concentration and the atomic scale spatial distribution—collectively reinforcing the critical role of boron impurities. The inhomogeneous stress condition in the sample was revealed by atomic-resolution STEM-EELS measurement, and no obvious impact should be imposed on the thermal measurement. This work advances the fundamental understanding of phonon scattering mechanisms in semiconductors and provides critical guidance for thermal management of next-generation semiconductor devices.

IV. MATERIALS AND METHODS

A. 3C-SiC single crystal growth

The single-crystal 3C-SiC samples studied in this work are grown by Air Water Inc. The growth process is performed on a (111) Si substrate at 1300 K via LT-CVD in a custom-designed CVD reactor. Both the growth temperature and Si substrate orientation are carefully optimized to enable high-quality single-crystalline 3C-SiC epitaxial growth. These 3C-SiC single crystals are expected to have low crystal defect densities, including stacking faults and double positioning boundaries at relatively low growth temperatures due to the shared rotational symmetry (120°) between Si and 3C-SiC around the [111] axis. The 3C-SiC films are then polished to a final thickness of ~3.5 μm (thermally thick regime) with a smooth surface for TDTR measurements.

B. Material characterizations by Raman spectroscopy and EBSD

Raman spectroscopy was performed using a Horiba LabRAM confocal Raman imaging system. The acquisition time is 60s over 5 cycles. The laser wavelength is 532 nm. The objective is 50x, and the total energy setting is 5%. The crystal orientation maps of the sample surfaces are obtained using EBSD measurement. A high-resolution Hitachi SU5000 scanning electron microscope (SEM) generates backscattered electrons containing crystallographic information, which are detected by an EDAX Hikari CCD camera. The scanning process is performed over a total area of 500×500 μm^2 with a step size of 3 μm. The EBSD measurements are conducted at WinTech Nano-Technology Services Pte Ltd.

C. Dopant characterization by SIMS and APT

The precise depth profiles of doping elements, including B, N, and O, within the crystals were determined using SIMS. A device from Cameca, Inc. accelerated C_s^+ to 10 keV to detect N and O, and a beam of accelerated at 12.5 keV was used to measure the concentration of B. The SIMS characterizations were conducted at Evans Material Technology (Shanghai) Co., Ltd. The APT specimens were prepared by Ga FIB milling. The APT measurements were performed with a local electrode atom probe (LEAP 5000XR) under UV laser (355 nm) pulsing at a pulse energy of 80 pJ, a pulse repetition rate of 200 kHz, and a target evaporation rate of 0.5% per pulse at 40 K. The reconstruction and quantitative analysis of APT data were performed using a CAMECA visualization and analysis software (AP Suite 6.3). The APT measurement is conducted at Shanghai Fuxin Technology Consulting Co., Ltd.

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D. Electron microscope characterization

High-resolution TEM and STEM were conducted by a spherical aberration-corrected FEI Titan Cubed Themis G2 300 microscope. Samples were fabricated by a Thermo Fisher Helios G4 UX FIB system. Atomic-resolution STEM-EELS data were acquired by a Nion U-HERMES200 micoscope at an electron accelerating voltage of 60 kV. The convergence semiangle was 35 mrad, and the collection semiangle was 25 mrad. For each data point, the EELS camera exposure time was 15 s to increase the signal-to-noise ratio. When the beam was positioned onto the sample, the energy resolution was around 15 meV and gradually decreased to about 17 meV near the Si/3C-SiC interface region due to the increasing sample thickness and higher defect density, as shown in Fig. S5.

E. EELS data processing

For each data point, the zero-loss peak (ZLP) was first aligned. The ZLP was then removed by fitting a Pearson VII function to two distinct energy windows before and after the phonon energy range of 3C-SiC: 25–35 meV (before) and 200–240 meV (after). The ranges of these fitting windows were dynamically adjusted across data points to optimize the fitting performance. Subsequently, residual phonon signals were smoothed using a Gaussian filter to suppress Gaussian noise. A detailed description of the signal processing workflow is provided in the supplementary material.

F. Thermal characterization by TDTR

The thermal conductivity of the 3C-SiC crystals was measured using TDTR, a non-contact optical technique based on a femtosecond laser source and pump-probe interaction. In TDTR, the pump beam is modulated at a specific frequency and focused onto the sample through an objective lens, inducing periodic temperature fluctuations. Meanwhile, a mechanical delay stage adjusts the optical path of the pump beam to control the delay time between the arrivals of the pump and probe beams. The probe beam, focused by the same objective lens, detects surface temperature variations by thermoreflectance. To enhance the thermoreflectance signal, a ~85 nm aluminum (Al) transducer layer was deposited on the sample surface. The reflected probe beam is detected by a silicon photodiode, and a lock-in amplifier extracts the signal component at the modulation frequency. By fitting the measured signal to an analytical solution of the sample structure, the unknown thermal properties—specifically, the thermal conductivity of 3C-SiC and the thermal boundary conductance (TBC) of the Al/3C-SiC interface—can be determined. In this study, the pump and probe beam radii were both 12 μm, with a modulation frequency of 10.1 MHz. The sensitivity analysis can be found in Fig. S9.

G. Picosecond acoustic technique

An ultrafast laser pulse train generates coherent strain waves that propagate through the thin-film layers at the longitudinal phonon group velocity, reflecting independently at each material interface. The reflected strain waves modulate the optical reflectivity of the Al transducer surface, which is subsequently detected by the probe beam at progressive time delays. These modulations appear as peaks or valleys in the in-phase signal, determined by the acoustic impedance Z contrast between adjacent materials at each interface. Here $Z = \rho v$, where ρ is the density of the layer and v is the sound speed. The round trip propagation time of strain waves through individual layers is obtained from the temporal separation between corresponding echo features. Using this approach, the thickness of each film can be precisely determined at the local measurement spot.

SUPPLEMENTARY MATERIAL

See the supplementary material for detailed information about the SIMS, EBSD, APT, EELS, and TDTR measurements.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Zifeng Huang: Data curation (equal); Formal analysis (equal); Investigation (lead); Methodology (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Jianbo Liang:** Conceptualization (equal); Methodology (equal); Writing – original draft (supporting); Writing – review & editing (supporting). **Yuxiang Wang:** Methodology (equal); Writing – original draft (supporting); Writing – review & editing (supporting). **Zixuan Sun:** Methodology (equal); Writing – original draft (supporting); Writing – review & editing (supporting). **Naoteru Shigekawa:** Conceptualization (equal); Methodology (equal); Writing – original draft (supporting); Writing – review & editing (supporting). **Ming Li:** Writing – original draft (supporting); Writing – review & editing (supporting). **Runsheng Wang:** Writing – original draft (supporting); Writing – review & editing (supporting). **Zhe Cheng:** Conceptualization (lead); Investigation (supporting); Project administration (lead); Software (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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